Bifurcation-Guided Replay: Learning at the Success–Failure Boundary of Decision Policies

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Abstract

Policies trained on nominal starts often fail under small deviations from the states encountered during training. Existing curriculum and level-replay methods prioritize difficult or high-regret levels, but they do not explicitly identify where a policy's behavior transitions from likely success to likely failure. We introduce Bifurcation-Guided Replay (BGR), a general replay method for sequential decision policies. For each replayable decision state, BGR estimates a recovery curve measuring success probability under perturbations of increasing scale, derives a critical radius at which performance crosses a success threshold, and samples training perturbations near this state-conditioned boundary. To make boundary estimation practical, BGR uses active probing, monotone curve fitting, uncertainty-aware refresh, and diversity-preserving replay. We evaluate BGR first in a diagnostic recovery-margin benchmark and then extend the same machinery to procedural decision tasks and robot suffix recovery. BGR improves recovery AUC and shifts critical-radius distributions toward larger margins compared with uniform replay, fixed-radius perturbation training, and failure-focused replay, while preserving clean success.

1 Introduction

A policy can solve a task and still be brittle: a minor shift in object pose, grid layout, browser state, or trajectory prefix may move it outside its recovery basin. Standard success metrics hide this geometry. We ask a different question: for each replayable state, how far can the current policy be perturbed before success becomes failure?

Existing replay and curriculum methods rank examples by difficulty, temporal-difference error, regret, novelty, or failure. These signals are useful, but they do not explicitly model the local response curve around a state. BGR instead estimates a state-conditioned recovery curve and trains near its success–failure transition.

Our contributions are:

- a domain-general replay formalism based on replayable decision states, perturbation families, recovery curves, critical radii, and recovery AUC;
- an efficient BGR algorithm that actively estimates success–failure boundaries and samples both states and perturbation radii near those boundaries;
- an active-estimation validation showing that boundary-focused probes recover critical radii with small fixed budgets rather than dense sweeps;
- a diagnostic benchmark and experiment pipeline for measuring recovery-margin expansion under controlled replay strategies;
- a path to procedural RL and robot suffix-recovery experiments using the same estimator and priority interfaces.

2 Related Work

BGR is closest in spirit to curriculum learning and level replay for reinforcement learning. Prioritized Level Replay samples levels by learning-potential proxies such as value loss (Jiang, Grefenstette, and Rocktäschel 2021), while PAIRED and ACCEL generate or mutate environments using regret and curriculum-frontier objectives (Dennis et al. 2020; Parker-Holder et al. 2022). BGR is complementary: it can operate over fixed replayable states and prioritizes the local perturbation geometry of the current policy rather than a scalar difficulty or regret score. It also differs from broad curriculum schedules surveyed by Narvekar et al. (2020), because the unit of progress is recovery-margin expansion around each state.

BGR also connects to robustness training and robotics generalization. Domain randomization improves transfer by broadening training distributions (Tobin et al. 2017), but fixed perturbation distributions can oversample states that are already easy or currently infeasible. BGR instead estimates where each state's perturbation curve crosses from success to failure and trains near that radius. Our procedural experiments use resettable grid states in the style of lightweight grid benchmarks (Chevalier-Boisvert et al. 2023); the suffix study is a diagnostic step toward manipulation settings such as LIBERO (Liu et al. 2023) and vision-language-action policies such as OpenVLA (Kim et al. 2024).

3 Problem Setting

Let π_θ be a sequential decision policy and let $Y(\tau) \in \{0, 1\}$ denote terminal success for rollout τ. A replayable decision

Bifurcation-Guided Replay
Initialize replay buffer $\mathcal{B}$ with replayable states.
For each $x \in \mathcal{B}$, probe $\sigma \in \{0, \sigma_{\min}, \sigma_{\text{mid}}, \sigma_{\max}\}$, fit monotone $\hat{R}(x, \sigma)$, and estimate $\hat{r}_\alpha(x)$.
For each training iteration, sample $x \sim q_{\mathcal{B}}(x)$ proportional to BGR priority, sample σ from a mixture centered at $\hat{r}_\alpha(x)$, apply $\delta \sim \mathcal{P}_x(\sigma)$, collect rollout or teacher recovery data, update π_θ, and periodically refresh stale curve estimates.

Figure 1: BGR training loop without domain-specific learner details.

state $x \in \mathcal{X}$ is any object from which evaluation or training can restart, such as an environment seed, simulator checkpoint, task prefix, or robot suffix state.

For each x, a domain-valid perturbation family applies $\tilde{x} = T(x, \delta)$ with $\delta \sim \mathcal{P}_x(\sigma)$, where $\sigma \geq 0$ is a perturbation magnitude. The recovery curve is

$$R_\pi(x, \sigma) = \Pr_{\delta, \tau}[Y(\tau) = 1 \mid \tau \sim \pi(\cdot \mid T(x, \delta))]. \tag{1}$$

For threshold α, the critical radius is

$$r_{\pi,\alpha}(x) = \sup\{\sigma : R_\pi(x, \sigma) \geq \alpha R_\pi(x, 0)\}. \tag{2}$$

We evaluate robustness with recovery AUC,

$$\text{RAUC}_\pi(x) = \frac{1}{\sigma_{\max}} \int_0^{\sigma_{\max}} R_\pi(x, \sigma) d\sigma. \tag{3}$$

4 Bifurcation-Guided Replay

BGR maintains a buffer of replayable states. For each state, it probes perturbation radii, fits a monotone recovery curve, estimates a critical radius, and assigns replay priority to states whose boundaries are feasible, sharp, uncertain, and near a trainable target band.

The priority for a record is a product of gates and soft utilities:

$$P(x) = G(x) U_{\text{boundary}}(x) U_{\text{sharp}}(x)^\lambda U_{\text{unc}}(x)^\beta U_{\text{stale}}(x)^\eta. \tag{4}$$

We mix this distribution with uniform replay to prevent starvation.

5 Experiments

All experiments are driven by checked-in YAML configs and scripts. The repository records per-seed CSVs, aggregation code, paired exact sign-flip tests, Slurm launch commands, remote log paths, compute/GPU environment snapshots, and an MIT license; the paper tables are generated from these artifacts rather than hand-edited results.

5.1 Active Boundary Estimation

BGR is useful only if recovery boundaries can be estimated without evaluating dense curves for every replay state. We therefore validate the training-time estimator separately from the replay experiments. For each synthetic state, we compute a dense 201-point reference curve for evaluation only, then compare three 17-probe estimators: uniform random radii, a repeated coarse grid, and active BGR probing initialized with a coarse sweep and then focused near the current critical-radius estimate.

Table 2 shows that active BGR finds the boundary within ± 0.08 radius units more often than uniform probing (0.682 vs. 0.615) and coarse probing (0.682 vs. 0.654). Its mean r_{80} error is close to the coarse sweep (0.079 vs. 0.072) while retaining the adaptive query mechanism used during replay. This result does not claim that active probing dominates every fixed grid in this simple one-dimensional setting; rather, it verifies that BGR's boundary estimates are usable at a small fixed rollout budget before final dense-curve evaluation.

5.2 Tier-0 Recovery-Margin Benchmark

The initial benchmark creates replayable states with latent feasible radii and policy recovery margins. Success probability follows a sigmoid recovery curve, and training updates near the current margin expand that margin most efficiently. This diagnostic environment tests whether BGR's estimator and sampler recover the intended behavior before expensive RL or robotics runs.

In the current three-seed diagnostic run, BGR improves final recovery AUC over uniform replay (0.3756 vs. 0.3659), improves the area under the RAUC learning curve (0.3169 vs. 0.2987), and preserves clean success (0.8962 vs. 0.8886). Fixed-radius, failure-only, and PLR-loss proxy baselines underperform in this controlled setting, suggesting that explicit boundary-radius sampling is useful when the benchmark contains learnable success–failure transitions.

5.3 Procedural Grid-Margin Study

We instantiate replayable states in procedural grid environments with resettable seeds and mid-path states. Generated obstacle maps define feasible perturbation neighborhoods through shortest-path reachability, while each replay state has a policy-specific recovery margin. This benchmark isolates whether BGR can expand state-conditioned recovery curves in a non-robotic sequential decision setting.

In a five-seed grid-margin run, BGR outperforms uniform replay on final RAUC (0.4350 vs. 0.3979), RAUC learning-curve area (0.3538 vs. 0.3145), median r_{80} (0.3442 vs. 0.3322), and clean success (0.9474 vs. 0.8969). Failure-only replay, PLR-loss proxy replay, and fixed-radius perturbation training substantially underperform, indicating that neither failure mining nor static perturbation scales recover the same margin-expansion behavior. Separate tabular grid-policy diagnostics either saturate under broad replay or undertrain BGR relative to fixed-radius oracle imitation. A coverage-aware BGR variant improves the policy diagnostic substantially over narrow BGR, but still trails broad fixed/loss-priority replay; we therefore treat this policy-level tabular setup as a negative setup result rather than evidence for the main claim.

Table 3 isolates the replay decisions inside BGR. The uncertainty and sharpness factors are not decisive in this benchmark, but the radius-level boundary rule is: keeping BGR state priority while sampling perturbation radii uniformly reduces RAUC from 0.435 to 0.385 and AULC from 0.354 to 0.307. This supports the central claim that BGR is not only

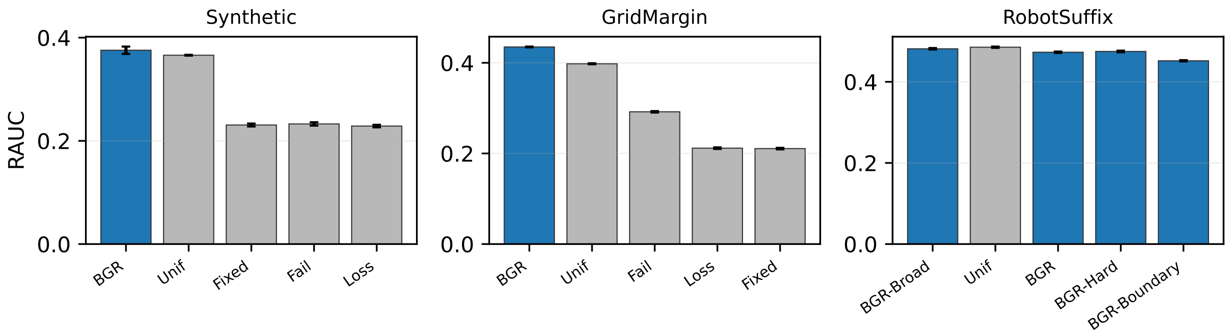

Figure 2: Final recovery AUC across the current diagnostic benchmarks. Bars show means over seeds and error bars show standard error. BGR improves RAUC in the synthetic and grid-margin studies; the suffix simulator exposes a remaining tradeoff where BGR-Suffix improves clean success and sample efficiency but not final object-perturbation RAUC relative to uniform suffix replay.

Benchmark	Method	Clean	RAUC	Median r_{80}	RAUC AULC
Synthetic margin	BGR	0.8962	0.3756	0.2829	0.3169
Synthetic margin	Uniform	0.8886	0.3659	0.2753	0.2987
Grid margin	BGR	0.9474	0.4350	0.3442	0.3538
Grid margin	Uniform	0.8969	0.3979	0.3322	0.3145
Robot suffix	BGR-Suffix Broad	0.8700	0.4815	0.4704	0.3798
Robot suffix	Uniform suffix	0.8368	0.4854	0.5032	0.3716

Table 1: Current diagnostic results, reported as seed means. Full per-seed CSVs and standard-error summaries are generated by the repository scripts. BGR is strongest in the synthetic and procedural grid-margin settings. In the lightweight robot-suffix simulator, a broad-radius BGR-Suffix variant improves clean success and sample efficiency over uniform suffix replay, but uniform still has slightly higher final object-perturbation RAUC.

Estimator	r_{80} MAE	RAUC MAE	Hit rate
Active BGR	0.079±0.001	0.066±0.001	0.682±0.004
Coarse	0.072±0.001	0.060±0.001	0.654±0.007
Uniform	0.106±0.002	0.065±0.000	0.615±0.007

Table 2: Probe-efficiency validation on synthetic recovery curves. All methods use 17 Bernoulli probes per state and are evaluated against dense 201-point curves. Active BGR improves boundary-hit rate over uniform and coarse probing, while keeping critical-radius error close to the coarse sweep.

hard-state prioritization; it also depends on training at the estimated success–failure radius.

5.4 Robot Suffix Study

BGR-Suffix treats a robot demonstration suffix as a replayable decision state. The current lightweight suffix simulator models language-conditioned manipulation families, demonstration suffix phase, object-pose perturbations, end-effector perturbations, teacher feasibility, and clean-demo regularization. Training uses object-pose offsets; end-effector offsets are used as a perturbation-family transfer diagnostic.

In a five-seed suffix run, the original boundary-heavy BGR-Suffix improves clean success over uniform suffix replay (0.8738 vs. 0.8368) and slightly improves RAUC learning-curve area (0.3745 vs. 0.3716), but uniform has higher final object-perturbation RAUC. A follow-up strategy comparison shows that the issue is radius coverage rather than state priority: a broad-radius BGR-Suffix variant reaches final object RAUC 0.4815, close to uniform's 0.4854, while retaining much higher clean success (0.8700 vs. 0.8368), higher end-effector transfer RAUC (0.3175 vs. 0.3081), and higher RAUC AULC (0.3798 vs. 0.3716). A hard-biased variant gives the best transfer and sample efficiency (transfer RAUC 0.3292, AULC 0.3865) but lower object RAUC. We therefore treat this suffix simulator as a diagnostic BGR-Suffix instantiation rather than final robotics evidence; the next step is a learned LIBERO/OpenVLA-style policy run with real task rollouts.

We also validated the real-simulator path for that next step. On five LIBERO-Goal tasks, three reset states per task, five object-perturbation radii, and four trials per radius, a GPU/EGL probe instantiated LIBERO environments, loaded benchmark init states, applied object free-joint perturbations up to 8cm, and completed zero-action rollouts without simulator errors (75/75 valid radius rows). In addition, we converted existing closed-loop OpenVLA/LIBERO object-task rollouts into BGR recovery curves over nine replay states. The fixed OpenVLA policy had clean success 1.0 on these states, but mean RAUC dropped under occlusion (0.389), blur (0.467), and image shift (0.515), with cor-

Method	Clean	RAUC	Median r_{80}	AULC
BGR	0.947±0.002	0.435±0.001	0.344±0.002	0.354±0.001
No uncertainty	0.947±0.001	0.436±0.000	0.343±0.001	0.354±0.001
No sharpness	0.948±0.002	0.436±0.001	0.345±0.002	0.354±0.001
Uniform radius	0.894±0.001	0.385±0.001	0.323±0.001	0.307±0.001
Uniform replay	0.897±0.001	0.398±0.001	0.332±0.001	0.315±0.001

Table 3: Grid-margin ablations over five seeds. Boundary-centered radius sampling is the important ingredient in this benchmark: replacing it with uniform radii drops RAUC below uniform replay, while removing uncertainty or sharpness weights has little effect.

responding mean r_{80} values 0.282, 0.307, and 0.379. A separate OpenVLA boundary-selection audit found proposal-guided perturbations in the observed counterfactual-failure band $[0.25, 0.75]$ more often than random-balanced selection (0.875 vs. 0.575). This is learned-policy evidence that recovery curves expose VLA brittleness and that boundary-seeking selection is measurable, but it is not yet a BGR fine-tuning result.

6 Limitations

BGR requires reset or replay access, a meaningful perturbation family, and extra rollouts for boundary estimation. It does not certify robustness, and the term bifurcation is used operationally for an observed success–failure transition rather than a formal dynamical-systems claim. The current OpenVLA result evaluates recovery curves for a fixed learned policy; a full robotics claim still requires BGR-Suffix fine-tuning and baselines on the same learned-policy protocol.

7 Conclusion

BGR converts recovery-margin measurement into a replay curriculum. By training at feasible success–failure boundaries, it targets the states and perturbation scales where policy improvement should most directly expand local recovery basins.

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